DOME SUNLIGHT COLLECTOR - GENERAL ARRANGEMENT

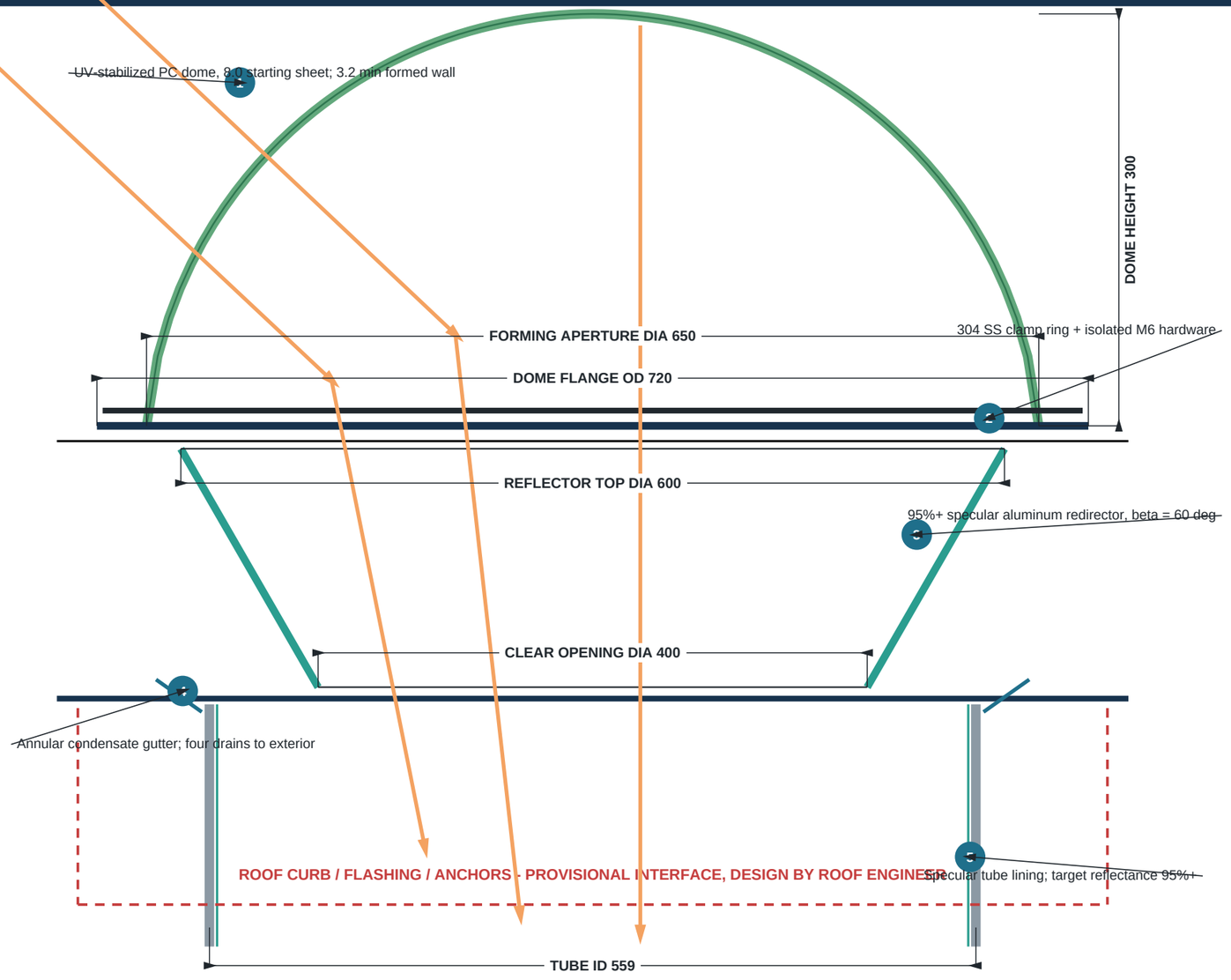
Roof-interface-neutral prototype; non-proprietary optical collector head for a nominal 559 mm tubular daylight device

UNITS: mm unless noted | SCALE: diagrammatic

ENGINEERING HOLD POINT - PERMANENT ROOF INSTALLATION IS NOT YET DEFINED

Build the collector head as a prototype only. Final curb, flashing, anchors and structure require roof type/slope, project location, adopted code, wind pressure and snow load.

SECTION A-A - COLLECTOR HEAD AND CONTROLLED ROOF INTERFACE



DESIGN BASIS - FIXED FOR PROTOTYPE

Tube clear diameter	559 (22 in nominal)
Dome geometry	650 aperture x 300 rise; spherical R326
Dome material	UV PC, 8.0 starting; 3.2 min formed
Collector base	304 SS, OD780 x 3; aperture DIA559
Optical redirector	DIA600 / DIA400 x H173.2; 60 deg
Seal concept	6 EPDM gasket + secondary 6 O-ring
Roof interface	PCD750 provision only - NOT RELEASED

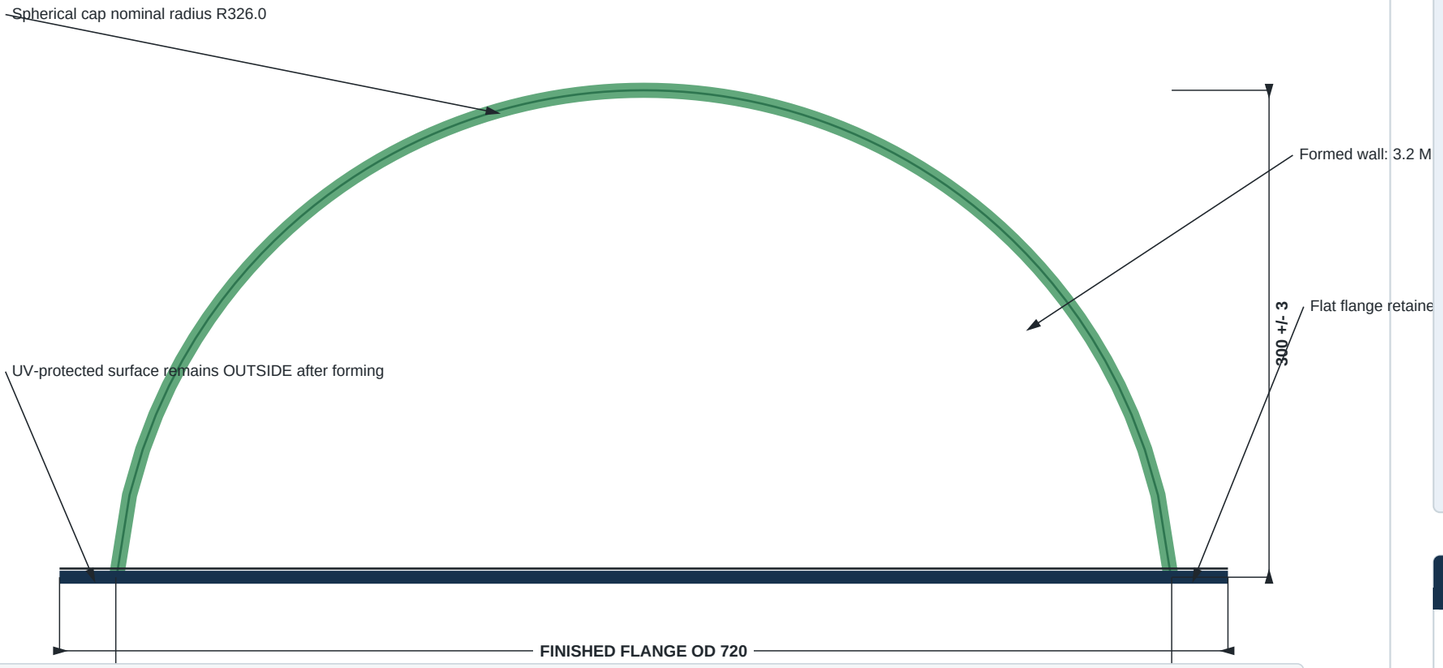
PROJECT INPUTS REQUIRED BEFORE ROOF RELEASE

Roof system / material / slope	TBD
Project location + adopted code edition	TBD
Design wind pressure: positive and negative	TBD
Ground and roof snow loads	TBD
Hail / impact classification	TBD
Fire / roof-covering classification	TBD
Curb height, insulation and flashing system	TBD

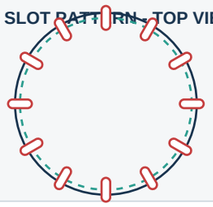
OPTICAL PRINCIPLE AND PERFORMANCE INTENT

1. The clear dome accepts diffuse sky light and direct sun over a wide azimuth range.
2. The annular 60 deg mirror redirects a 30 deg solar-altitude ray approximately vertical.
3. The DIA400 central opening passes high-angle sunlight directly into the tube.
4. For solar altitude alpha, reflected deviation from vertical is approximately alpha - 30 deg.
5. Optical output is a design intent only; verify with lux mapping at 15, 30 and 60 deg incidence.

PART D-01 - FORMED POLYCARBONATE DOME (SECTION)



FLANGE SLOT PATTERN - TOP VIEW (schematic)



12 radial slots 8 x 18 equally spaced at 30 deg on PCD680

Slot long axis radial; deburr and polish edges; no sharp internal corners

Slots accommodate about 1.9 mm radial growth for Delta-T = 80 C

FORMED-WALL MATERIAL BUDGET - WHY THE STARTING SHEET IS 8 mm

Spherical cap area = 2 x pi x R x h = 2 x pi x 326.0 x 300 = 614,500 mm2 (nominal)

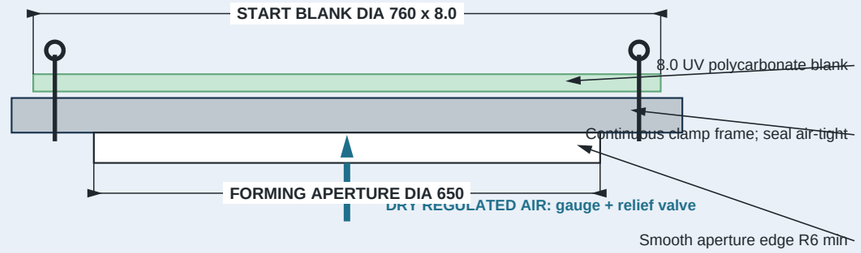
Forming aperture area = pi x 325 x 325 = 331,800 mm2 | draw ratio = 614,500 / 331,800 = 1.852

Ideal average cap wall from 8.0 sheet = 8.0 / 1.852 = 4.32 mm before local thinning and process variation.

A 5.0 mm sheet would average only 2.70 mm, so it cannot reliably meet a 3.2 mm minimum formed wall.

This calculation does not replace thickness mapping. Reject any part with a reading below 3.2 mm.

STARTING BLANK AND FREE-BLOW FIXTURE



CONTROL	PROTOTYPE SETTING / ACCEPTANCE
Material	UV-stabilized optical PC; protected both sides
Pre-dry 8 mm	125 +/- 3 C for 36 h; follow sheet supplier
Forming window	170-180 C suggested for dome forming trials
Pressure	Trial-determined; regulator, gauge and relief required
Height stop	300 +/- 3; calibrated non-contact gauge
Release	Hold shape until rigid; about 120-125 C part temp

DOME QA, TRIM AND HANDLING

1. Mark a 50 x 50 inspection grid over the cap.
2. Map ultrasonic or mechanical thickness at every node.
3. Accept only if every cap reading is 3.2 mm or greater.
4. Trim finished flange to DIA720 concentric within 1.0 TIR.
5. Machine 12 slots 8 x 18 on PCD680; edge radius 1 min.
6. Inspect for haze, bubbles, webbing, scratches and UV-side reversal.
7. Do not flame-polish PC; avoid stress-cracking cleaners/sealants.
8. Record sheet lot, heat cycle, height, pressure history and thickness map.

PROCESS SAFETY

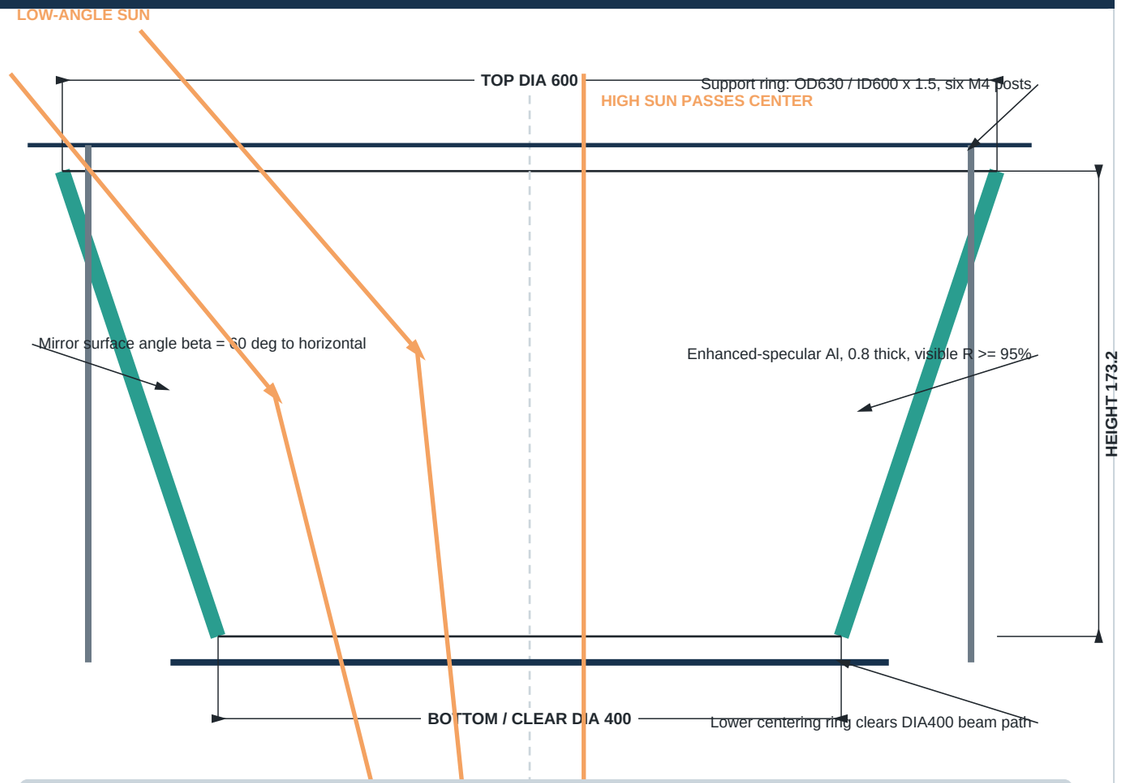
Heated sheet and compressed air store significant energy. Fixture, pressure limits, guarding and operator procedures must be engineered by the forming shop.

OPTICAL REDIRECTOR - CONE, FLAT PATTERN AND RAY DESIGN

Annular enhanced-specular aluminum insert; generic geometry optimized for 30 deg solar altitude

UNITS: mm unless noted | SCALE: diagrammatic

PART O-01 - ANNULAR CONICAL REFLECTOR

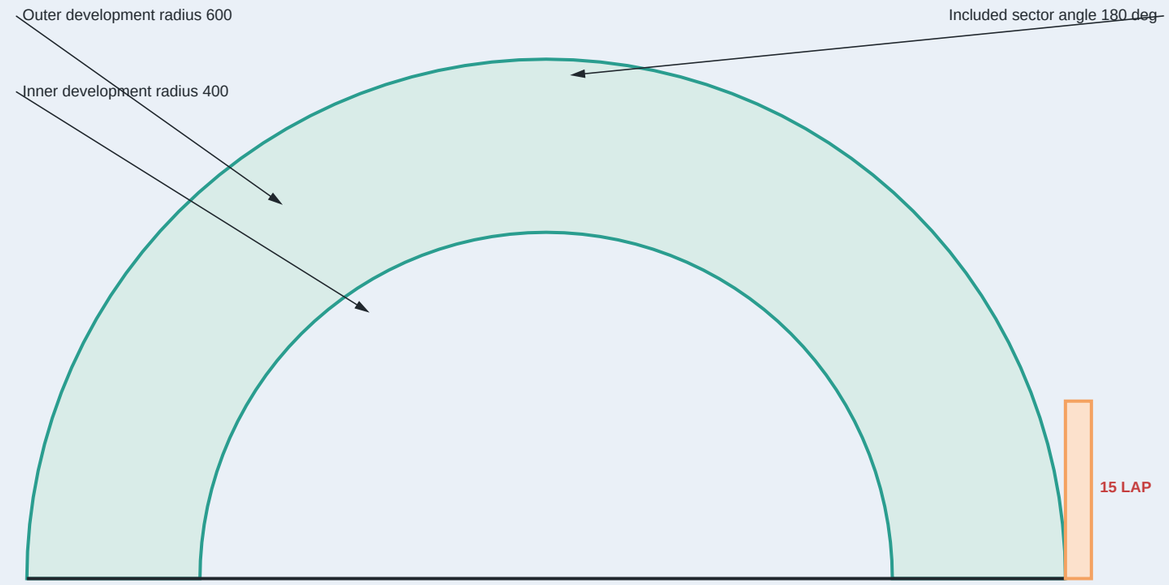


Finished slant length: 200.0 | radial change: 100.0 | axial height: sqrt(200^2 - 100^2) = 173.2

Roll with protective film on optical face. Seam at north/reference side; 15 lap, flush rivet or bonded lap.

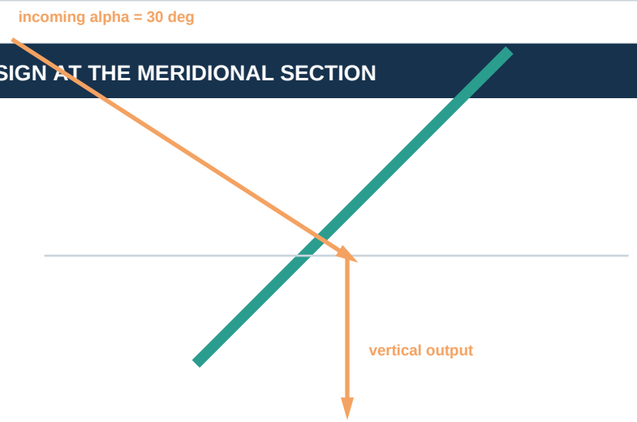
No fastener head may protrude into the mirror face or the DIA400 central clear opening.

FLAT PATTERN O-01 - HALF-ANNULAR SECTOR



CUT SIZE derives from cone radii 300/200 and slant length 200. Verify a paper template before cutting optical aluminum.

RAY DESIGN AT THE MERIDIONAL SECTION



SPECULAR RELATION

beta = 45 deg + alpha0 / 2 = 60 deg for alpha0 = 30 deg

Approximate reflected deviation from vertical: delta = alpha - alpha0

alpha = 15 deg -> delta = -15 deg | alpha = 30 deg -> 0 deg | alpha = 60 deg -> +30 deg

This 2D law sets the cone slope. Full 3D performance varies with solar azimuth, dome refraction, surface errors and tube losses.

Acceptance is photometric: lux map at the diffuser for 15, 30 and 60 deg elevation and multiple azimuths.

REDIRECTOR FABRICATION CONTROLS

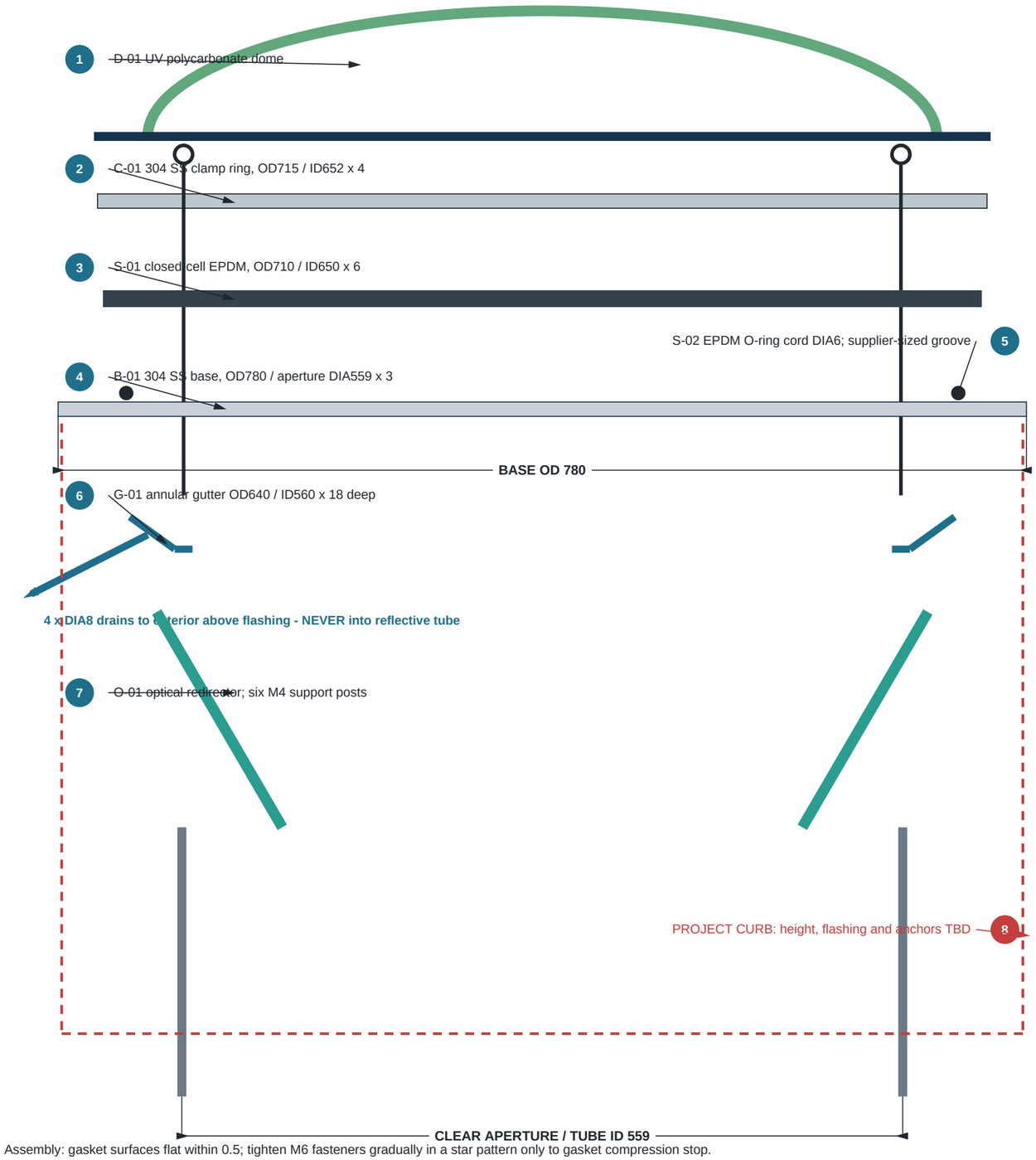
1. Optical-face reflectance certificate: 95% visible or better.
2. Cut half-annulus R600/R400, 180 deg; add 15 lap.
3. Roll without scratching; use soft covered rolls and gloves.
4. Join seam behind lap; keep inner face smooth and clean.
5. Finished DIA600 / DIA400 / H173.2, each +/- 1.0.
6. Total runout at support ring: 1.5 max.
7. Clean only with material-approved non-abrasive method.
8. Orient seam to roof reference mark for repeatable testing.

BASE, SEALS, CONDENSATE DRAINAGE AND ROOF INTERFACE

Exploded assembly and controlled interface dimensions; curb and anchorage remain project-specific

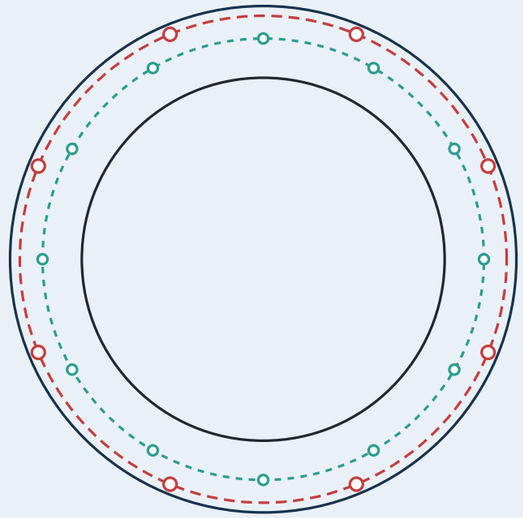
UNITS: mm unless noted | SCALE: diagrammatic

EXPLODED SECTION - ASSEMBLY ORDER



Assembly: gasket surfaces flat within 0.5; tighten M6 fasteners gradually in a star pattern only to gasket compression stop.

BASE PLATE B-01 - TOP VIEW AND HOLE CONTROL



ROOF HOLES: HOLD

Do not drill PCD750 curb holes until curb, anchor edge distance, wind uplift and flashing are approved.

DETAIL S-1 - DOME CLAMP AND THERMAL MOVEMENT



Allow radial sliding; do not rigidly pin PC

BOM - COLLECTOR HEAD

ITEM	QTY	DESCRIPTION
D-01	1	Formed UV PC dome, 8 start / 3.2 min
C-01	1	304 SS clamp ring OD715/ID652 x 4
S-01	1	Closed-cell EPDM OD710/ID650 x 6
S-02	1	EPDM O-ring cord DIA6, 60A
B-01	1	304 SS base OD780/aperture559 x 3
G-01	1	SS gutter OD640/ID560 x 18 deep
O-01	1	Specular Al cone 600/400 x H173.2
H-01	12	M6 x 25 316 SS + bonded EPDM isolation
H-02	6	M4 SS reflector posts, H173 nominal

FABRICATION, ASSEMBLY, TEST AND RELEASE CHECKLIST

Prototype workflow and evidence required before any permanent roof installation

UNITS: mm unless noted | SCALE: diagrammatic

A. MANUFACTURE AND ASSEMBLE THE COLLECTOR HEAD

01	**PROCURE**	UV-stabilized optical PC 8.0; 304 SS plate/ring; 95%+ specular Al; compatible EPDM.
02	**FORM DOME**	Pre-dry, heat, free-blow through DIA650 fixture to 300 height; hold to rigid; retain process record.
03	**INSPECT DOME**	50 mm grid thickness map; reject below 3.2; visual check; trim OD720 and machine twelve 8 x 18 slots.
04	**MAKE REFLECTOR**	Cut R600/R400 half-annulus, 180 deg + 15 lap; roll to DIA600/DIA400 x H173.2.
05	**MAKE METALWORK**	Cut base OD780/aperture559, clamp ring OD715/ID652, support rings and gutter; deburr/passivate.
06	**DRY ASSEMBLE**	Install reflector and gutter. Check DIA400 clear path, drainage fall, runout and tube alignment.
07	**SEAL DOME**	Fit O-ring and EPDM gasket; place dome; install isolated M6 fasteners; tighten in star sequence.
08	**BENCH TEST**	Water, air, pressure, thermal cycle and optical tests. Record results and corrective actions.
09	**ROOF DESIGN**	Obtain roof/site loads and code. Engineer curb, anchors, insulation and flashing for the project.
10	**RELEASE**	Issue project-specific stamped/approved installation set only after structural and weather tests pass.

B. VERIFICATION MATRIX - DOCUMENT EVERY RESULT

TEST / INSPECTION	PROTOTYPE ACCEPTANCE	STATUS
Dome dimensions	OD720; cap DIA650; H300 +/- 3	OPEN
Dome thickness	50 grid; every point >= 3.2	OPEN
Visual / optical	No haze, webbing, bubbles or deep scratches	OPEN
Drainage	All 4 drains clear; no water enters tube	OPEN
Air leakage	ASTM E283 / adopted NAFS criterion	OPEN
Water penetration	ASTM E331 / adopted NAFS criterion	OPEN
Structural pressure	ASTM E330 at project design pressures	HOLD
Impact / hail	Classification required by adopted code	HOLD
Fire / roof covering	Classification required by adopted code	HOLD
Thermal cycling	No cracks, seal loss or fastener binding	OPEN
UV exposure	Supplier qualification + project requirement	OPEN
Photometric	Lux maps at 15/30/60 deg, several azimuths	OPEN
Roof curb / anchors	Engineered for roof and site loads	HOLD

C. PROJECT-SPECIFIC RELEASE DATA - ALL FIELDS REQUIRED

Project / address TBD

Design wind pressure + / - TBD

Roof material / assembly TBD

Ground / roof snow load TBD

Roof slope and curb height TBD

Hail and fire classification TBD

Adopted building code / edition TBD

Engineer / approval reference TBD

NO BUILDING INSTALLATION RELEASE UNTIL ALL FIELDS AND HOLD TESTS ARE CLOSED

D. TECHNICAL BASIS AND LIMITS

[1] POLYVANTIS, LEXAN Sheet Processing, Forming and Fabricating Guide, 2025 - drying, dome-forming and thermoforming guidance.

[2] POLYVANTIS, LEXAN XL102UV Sheet Datasheet - UV protection, transmission, thermal and forming properties.

[3] Public VELUX tubular unit skylight specification - nominal 559 mm unit and 3.18 mm minimum glazing reference.

[4] Alanod daylight-system materials - enhanced-specular aluminum reflectance capability up to about 98%.

[5] FGIA NAFS overview - tubular daylight device product classification and performance testing framework.

[6] Exact material grade, adopted code, NAFS edition, curb, flashing, anchors and load ratings remain project-specific.

polyvantis.com - LEXAN Sheet Processing, Forming and Fabricating Guide 2025
polyvantis.com - LEXAN XL102UV Sheet Datasheet, Americas
commercial.veluxusa.com/resources/tubular | alanod.com/applications/lighting-industry/daylight-systems
fgiaonline.org/resources/technical-resources/nafs-overview